RAG Architecture & Document Intelligence Guide

1. Executive Summary

Retrieval-Augmented Generation (RAG) significantly improves LLM responses by conditioning generation on authoritative local document context. In enterprise environments, document intelligence systems must handle heterogeneous documents (PDFs, multi-column layouts, tables, and scanned receipts) while maintaining strict citation grounding and low latency.

2. Document Ingestion Pipeline

The document ingestion process begins with structural document parsing. Using PyMuPDF for native vector text and pdfplumber for coordinate-aware table boundary detection, documents are decomposed into semantic units. Scanned pages or rasterized diagrams trigger optical character recognition (OCR) via Tesseract with image preprocessing.

3. Chunking Strategies and Granularity

Fixed character chunking often fractures important context. To mitigate this, recursive character splitting uses structural delimiters (headings, double newlines, sentence terminators). Chunks are targeted at 800 to 1200 characters with a 150-character sliding overlap to ensure boundary continuity across chunk transitions.

4. Hybrid Representation: Dense Vector & Sparse Lexical

Each text chunk receives dual representations:

- Dense Embeddings: Generated using BAAI/bge-small-en-v1.5 and indexed in PostgreSQL using pgvector (HNSW / IVFFlat).
- Sparse Inverted Index: Computed using PostgreSQL to_tsvector and tsquery for exact keyword, acronym, and entity recall.

5. Reciprocal Rank Fusion (RRF) & Re-ranking
Because dense similarity scores (cosine distance) and sparse BM25 scores (term frequencies) operate on disparate numerical scales, Reciprocal Rank Fusion (RRF) combines ranked lists without manual weight tuning:
RRF_score(d) = sum(1.0 / (k + rank_i(d))) where k is set to 60.

Top candidates from RRF fusion then undergo cross-encoder re-ranking using a cross-attention transformer (e.g. ms-marco-MiniLM-L-6-v2). The cross-encoder jointly computes token interactions between query and candidate chunks, filtering out false positives.

6. Agentic Orchestration with LangGraph
A stateful LangGraph workflow routes queries dynamically:
- Direct Answer: Fact-based questions with high retrieval confidence.
- Query Rewriting: Reformulates ambiguous or short queries into targeted retrieval keys.
- Table Analytics Mode: Directly routes questions about numerical aggregations to the deterministic tabular compute engine.
- Citation Alignment: Every claim maps back to verifiable page numbers and chunk hashes for strict verification.